

# Quantitative Evaluation of Localised Fibre Loss in Multicore Fibre Systems Using Coherent OTDR

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**Abstract** We quantitatively evaluate localised fibre loss in two-core fibre systems using coherent OTDR. Better than  $\pm 1$ -dB agreement with power meter measurements is confirmed for two distinct configurations, with and without high-loss loopbacks. Operational advantages and disadvantages of each configuration are also clarified. ©2026 The Author(s)

## Introduction

Coherent optical time-domain reflectometry (C-OTDR) is a primary technique for the reliable operation and maintenance of optical submarine cables, as it enables the localisation of cable faults by analysing Rayleigh backscattered light [1–12]. Research into C-OTDR for multicore fibre systems is progressing, with reports demonstrating the detection of cable break, repeater gain reduction, and localised fibre loss in configurations with high-loss loopbacks (HLLBs) [13, 14] and another that utilises inter-core crosstalk (IC-XT) instead of HLLBs to return the probe pulse to the C-OTDR [14, 15]. In particular, tracking temporal changes in localised fibre loss through in-service C-OTDR measurements is a key advantage, as it enables proactive cable repairs. Accurately quantifying localised fibre loss also allows for system-wide quality assurance after cable installation and repair. However, for both configurations, the quantitative evaluation of localised fibre loss has not been reported.

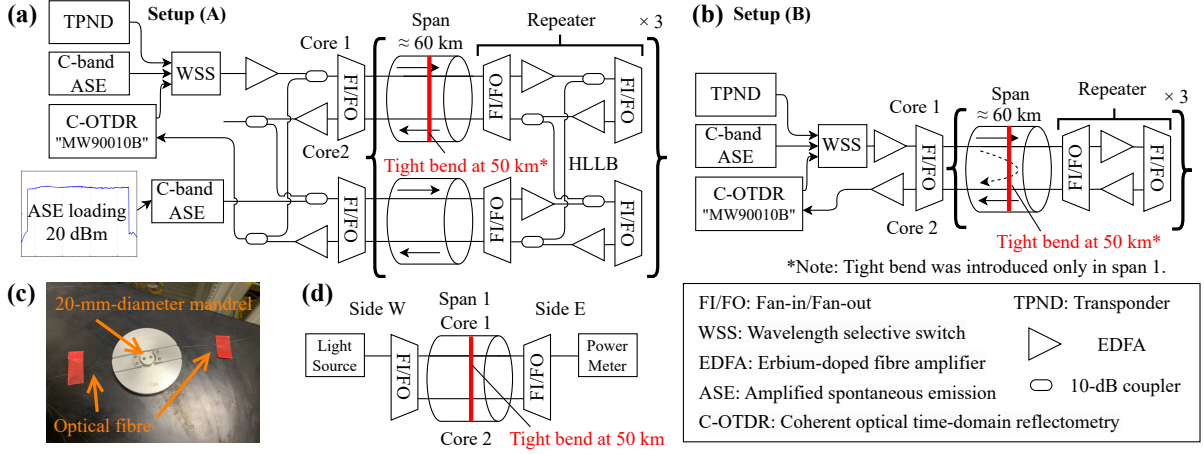
In this paper, we quantitatively evaluate the C-OTDR performance for localised fibre loss measurement in two configurations: one with HLLBs and another utilising IC-XT. We demonstrate that both configurations achieve better than  $\pm 1$ -dB agreement with power meter measurements. Furthermore, we clarify the advantages and disadvantages of each configuration, including the technical challenges in this quantitative measurement.

## Experimental setup

Based on our previous work [14], we configured two distinct experimental setups employing two-core fibres (2CFs). Figure 1(a) shows the experimental setup for the bidirectional system with two strands of 2CF, hereafter referred to as Setup (A). The probe pulse from the C-OTDR (MW90010B) was multiplexed in a wavelength selective switch with both a full C-band amplified spontaneous emission loading, which emulates data traffic, and a signal from a real-time transponder. The

combined signals were amplified to 20 dBm and launched into core 1 of span 1. Each of the three spans consisted of an approximately 60-km-long 2CF and a pair of fan-in/fan-out (FI/FO) devices. The backscattered light from the probe pulse was directed back to the C-OTDR through the HLLBs in each repeater. Figure 1(b) shows the experimental setup for the bidirectional system with a single strand of 2CF, where the repeaters are not equipped with HLLBs, hereafter referred to as Setup (B). The probe pulse returns to the C-OTDR by coupling to core 2 via IC-XT. For the C-OTDR measurements in both setups, the input power and wavelength of the probe pulse were set to 8 dBm and 1564.27 nm, respectively, for in-service measurement conditions [14]. The probe pulse width was set to 30  $\mu$ s, corresponding to a spatial resolution of approximately 3 km, and the number of averaging processes was  $2^{18}$ .

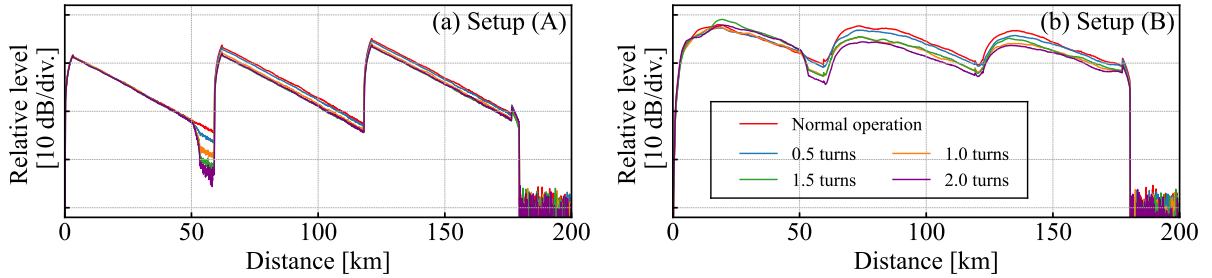
Localised fibre loss was introduced at the 50 km point of span 1 by wrapping the fibre around a mandrel with a diameter of 20 mm, as illustrated in Fig. 1(c). The 20-mm bend diameter is significantly smaller than the design specification for this fibre, which is compliant with the ITU-T G.654.B standard [16]. The loss was evaluated for four distinct numbers of turns: 0.5, 1.0, 1.5, and 2.0. Figure 1(d) illustrates the measurement configuration for evaluating the loss with an optical power meter, which consisted of span 1 connected between a pair of FI/FO devices. The measurements were performed for both core 1 and core 2 from each side of span 1, designated as Side W and Side E. Table 1 summarises the loss values, defined as the difference in optical power measured with and without the tight bend. Here,  $N$  represents the number of turns, while  $L_{1WE}$ ,  $L_{2WE}$ ,  $L_{1EW}$ , and  $L_{2EW}$  denote the loss values for the four respective measurements. Although the measurement configuration with a power meter was identical for both Setup (A) and Setup (B), the loss measurements were taken at different times to correspond with the acquisition of each C-OTDR trace. The



**Fig. 1:** Experimental setups for the C-OTDR measurements: (a) bidirectional system using two strands of two-core fibres with HLLBs and (b) bidirectional system using a single strand of two-core fibre without HLLBs. (c) A 20-mm-diameter mandrel for localised fibre loss at 50 km in span 1. (d) Measurement configuration using a power meter.

**Tab. 1:** Localised fibre loss from tight bend, measured with a power meter for each setup, where  $N$  represents the number of turns.

| Direction                              | Setup (A) |           |           |           | Setup (B) |           |           |           |
|----------------------------------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
|                                        | $N = 0.5$ | $N = 1.0$ | $N = 1.5$ | $N = 2.0$ | $N = 0.5$ | $N = 1.0$ | $N = 1.5$ | $N = 2.0$ |
| Core 1 W $\rightarrow$ E ( $L_{1WE}$ ) | 1.15      | 2.25      | 3.45      | 4.56      | 0.83      | 2.47      | 3.40      | 4.92      |
| Core 2 W $\rightarrow$ E ( $L_{2WE}$ ) | 0.18      | 0.67      | 1.51      | 1.37      | 0.36      | 0.58      | 0.76      | 1.02      |
| Core 1 E $\rightarrow$ W ( $L_{1EW}$ ) | 1.07      | 2.40      | 3.95      | 4.29      | 1.01      | 2.81      | 3.91      | 4.88      |
| Core 2 E $\rightarrow$ W ( $L_{2EW}$ ) | 0.28      | 0.75      | 1.44      | 1.58      | 0.40      | 0.64      | 0.81      | 1.22      |



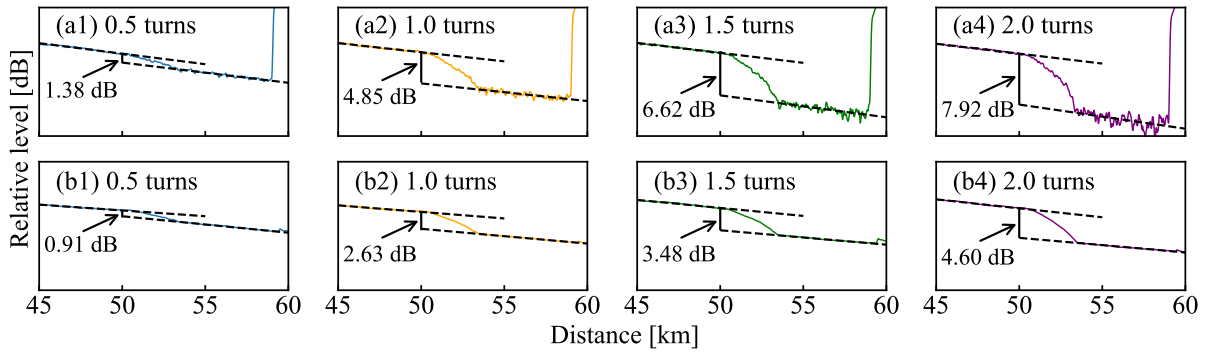
**Fig. 2:** C-OTDR traces under tight bend conditions with different numbers of turns in (a) Setup (A) and (b) Setup (B).

data in Tab. 1 indicate that the loss increases in proportion to the number of turns. A difference in loss is observable between the cores, which could be caused by the different refractive index profiles of the two cores.

### Detection of localised fibre loss

Figure 2(a) shows the C-OTDR traces corresponding to each level of localised fibre loss in Setup (A). The traces obtained through the HLLBs, which resemble those from conventional single-core fibre systems, enable the observation of the fibre loss profile for each span. At the 50 km point in the span 1, a drop in the trace level is evident, corresponding to the magnitude of the localised fibre loss. The noise level appears beyond the 180 km point, and the trace level subsequent to the loss point becomes more susceptible to noise as the number of turns increases. Figure 2(b) shows the C-OTDR traces in Setup (B). Although their overall shape differs from that in Setup (A) because the trace originates from IC-XT, the trace level still exhibits a drop corresponding to the magnitude

of the localised fibre loss. Figure 3 shows enlarged views of the traces at the loss points. The localised fibre loss was calculated from the vertical gap between two linear-fitting lines at the 50 km point. These lines were determined by applying the least-squares method to the trace data before and after the loss point, using the ranges 45.0–50.0 km and 53.5–59.0 km, respectively. Each range is longer than the spatial resolution of 3 km. The localised fibre loss measured in Setup (A) is consistently larger than that in Setup (B). This discrepancy arises from the difference in localised fibre loss between the cores, as shown in Tab. 1, and the distinct propagation paths of the probe pulse. In Setup (A), the probe pulse travels to and from the loss point in core 1 and returns to the C-OTDR via the HLLBs. Therefore, based on the loss values measured with a power meter, the total loss in Setup (A) is the sum of  $L_{1WE}$  and  $L_{1EW}$  from Tab. 1. Here, while this value represents the loss solely from core 1, the corresponding loss for core 2 is obtained by performing a C-OTDR measurement from the opposite direction. In contrast,



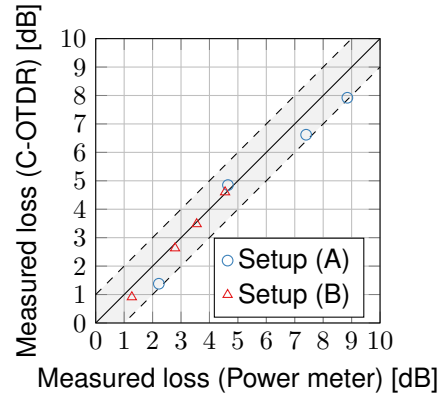
**Fig. 3:** Enlarged views of the C-OTDR traces at the loss points, with estimated loss values determined by linear fitting, where (a1)–(a4) show the results for Setup (A) and (b1)–(b4) for Setup (B).

**Tab. 2:** Advantages and disadvantages of each setup

|               | Setup (A)                                                                                            | Setup (B)                                                                 |
|---------------|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Advantages    | Fibre loss profile observable<br>IC-XT distribution observable<br>Loss distinguishable between cores | System simplicity<br>Reduced optical loss<br>No backscattering impairment |
| Disadvantages | Low compatibility with multicore amplifiers                                                          | Loss indistinguishable between cores<br>Longer near-end dead zone         |

Setup (B) involves two paths: (i) the pulse propagates round-trip through the loss point in core 1 before coupling to core 2 via IC-XT, and (ii) the pulse first couples to core 2 via IC-XT and propagates round-trip through the loss point in core 2. Any pulse component that couples to core 2 immediately after passing the loss point in core 1 is fundamentally negligible, as the power transferred over the infinitesimally short length is, by definition, minimal. Thus, the observable loss in Setup (B) is the average of the loss values from two paths (i) and (ii): the sum of  $L_{1WE}$  and  $L_{1EW}$ , and the sum of  $L_{2WE}$  and  $L_{2EW}$ . The average was calculated in the linear domain and then converted back to the dB scale. Figure 4 shows a comparison between these calculated results and the localised fibre loss estimated from the C-OTDR traces. The solid line indicates perfect agreement, while the dashed lines represent the  $\pm 1$ -dB error boundaries. As all data points lie within this shaded region, it confirms that the loss values estimated by the C-OTDR match the power meter measurements with an error within  $\pm 1$  dB in both setups.

Finally, we clarify the trade-offs that must be considered when choosing an appropriate configuration. Table 2 summarises the advantages and disadvantages of each setup. The use of HLLBs in Setup (A) is particularly advantageous as it provides more detailed link information. This approach, however, complicates integration with multicore amplifiers, as the HLLBs require two pairs of FI/FO devices around each amplifier. While Setup (B), a configuration eliminating HLLBs, offers several advantages, it also presents two technical challenges. The first challenge is the inability to distinguish the difference in loss between the individual cores. A severe loss occurring in a sin-



**Fig. 4:** Localised fibre loss measured with C-OTDR vs. with power meter, showing the  $\pm 1$ -dB error boundaries.

gle core could be masked by the averaging effect, appearing as a smaller loss on the C-OTDR trace and potentially leading to an incorrect assessment of the repair urgency. The second challenge is that the initial rise of the trace is more gradual compared to that in Setup (A), as seen in Fig. 2(b), because the trace originates from IC-XT. In this initial region, a localised fibre loss could not be quantified accurately by the linear-fitting method. Performing the C-OTDR measurement from the opposite direction could resolve this issue.

## Conclusions

We have quantitatively evaluated localised fibre loss using coherent OTDR in two different two-core fibre systems: one with high-loss loopbacks (HLLBs) and another without HLLBs. The experimental results revealed that both configurations achieve better than  $\pm 1$ -dB agreement with power meter measurements. The demonstrated measurement capability will enable proactive cable repair and quality assurance after cable installation and repair for multicore fibre submarine cables.

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## References

- [1] Y. Horiuchi, S. Ryu, K. Mochizuki, and H. Wakabayashi, "Novel Coherent Heterodyne Optical Time Domain Reflectometry for Fault Localization of Optical Amplifier Submarine Cable Systems", *IEEE Photonics Technology Letters*, vol. 2, no. 4, pp. 291–293, 1990. DOI: 10.1109/68.53266.
- [2] H. Izumita, Y. Koyamada, S. Furukawa, and I. Sankawa, "The Performance Limit of Coherent OTDR Enhanced with Optical Fiber Amplifiers due to Optical Nonlinear Phenomena", *Journal of lightwave technology*, vol. 12, no. 7, pp. 1230–1238, 1994. DOI: 10.1109/50.301816.
- [3] S. Furukawa, K. Tanaka, Y. Koyamada, and M. Sumida, "Enhanced Coherent OTDR for Long Span Optical Transmission Lines Containing Optical Fiber Amplifiers", *IEEE Photonics Technology Letters*, vol. 7, no. 5, pp. 540–542, 1995. DOI: 10.1109/68.384537.
- [4] M. Sumida, S. Furukawa, K. Tanaka, and M. Aiki, "High-Accurate Fault Location Technology Using FSK-ASK Probe Backscattering Reflectometry in Optical Amplifier Submarine Transmission Systems", *Journal of lightwave technology*, vol. 14, no. 10, pp. 2108–2116, 1996. DOI: 10.1109/50.541197.
- [5] H. Izumita, Y. Koyamada, S. Furukawa, and I. Sankawa, "Stochastic Amplitude Fluctuation in Coherent OTDR and a New Technique for Its Reduction by Stimulating Synchronous Optical Frequency Hopping", *Journal of lightwave technology*, vol. 15, no. 2, pp. 267–278, 1997. DOI: 10.1109/50.554377.
- [6] O. Gautheron, J. B. Leroy, and P. Marmier, "COTDR Performance Optimization for Amplified Transmission Systems", *IEEE Photonics Technology Letters*, vol. 9, no. 7, pp. 1041–1043, 1997. DOI: 10.1109/68.593393.
- [7] T. Otani, Y. Horiuchi, T. Kawazawa, K. Goto, and S. Akiba, "Fault Localization of Optical WDM Submarine Cable Networks Using Coherent-Optical Time-Domain Reflectometry", *IEEE Photonics Technology Letters*, vol. 10, no. 7, pp. 1000–1002, 1998. DOI: 10.1109/68.681297.
- [8] H. Iida, Y. Koshikiya, F. Ito, and K. Tanaka, "High-Sensitivity Coherent Optical Time Domain Reflectometry Employing Frequency-Division Multiplexing", *Journal of lightwave technology*, vol. 30, no. 8, pp. 1121–1126, 2012. DOI: 10.1109/JLT.2011.2170960.
- [9] Y. Song, L. Lu, and X. Zhang, "A Novel Multi-frequency Coherent OTDR for Fast Fading Noise Reduction", in *2012 Optical Fiber Communications Conference and Exhibition (OFC)*, Los Angeles, USA, 2012, OTu1C–4. DOI: 10.1364/OFC.2012.OTu1C.4.
- [10] R. Goldman, A. Agmon, and M. Nazarathy, "Direct Detection and Coherent Optical Time-Domain Reflectometry With Golay Complementary Codes", *Journal of lightwave technology*, vol. 31, no. 13, pp. 2207–2222, 2013. DOI: 10.1109/JLT.2013.2264273.
- [11] H. Iida, K. Toge, and F. Ito, "Environmental Perturbation Tracking in Coherent OTDR for Recovering Detection Sensitivity", in *39th European Conference and Exhibition on Optical Communications (ECOC 2013)*, London, UK, 2013, pp. 1–3. DOI: 10.1049/cp.2013.1385.
- [12] Y. Zhao, T. Yang, K. Yan, Z. Ren, and X. Chen, "High-performance and cost-effective C-OTDR employing polarization independent coherent detection", in *2018 Conference on Lasers and Electro-Optics (CLEO)*, San Jose, USA, 2018, pp. 1–2. DOI: 10.1364/CLEO\_AT.2018.JTh2A.186.
- [13] K. Komatsu et al., "Impact of Inter-Core Crosstalk on Coherent Optical Time-Domain Reflectometry in Repeated Multicore Fibre Systems", in *51th European Conference on Optical Communications (ECOC 2025)*, Copenhagen, Denmark, 2025. DOI: 10.1109/ECOC66593.2025.11263132.
- [14] K. Komatsu et al., "Repeater Fault Detection and Localization Using Coherent OTDR for Multicore Fiber Submarine Cable Systems", in *2026 Optical Fiber Communications Conference and Exhibition (OFC)*, Los Angeles, USA, 2026, Tu2F.4.
- [15] A. Nakamura, S. Ohno, and K. Toge, "Impact of Optical Loopback on Backward Crosstalk and Fault Localisation in Multi-Core Fiber Submarine Systems", in *51th European Conference on Optical Communications (ECOC 2025)*, Copenhagen, Denmark, 2025, p. M.03.06.5. DOI: 10.1109/ECOC66593.2025.11263135.
- [16] ITU-T. "Recommendation G.654: Characteristics of a cut-off shifted single-mode optical fibre and cable". [Online]. Available: <https://www.itu.int/rec/T-REC-G.654-202408-I/en>.